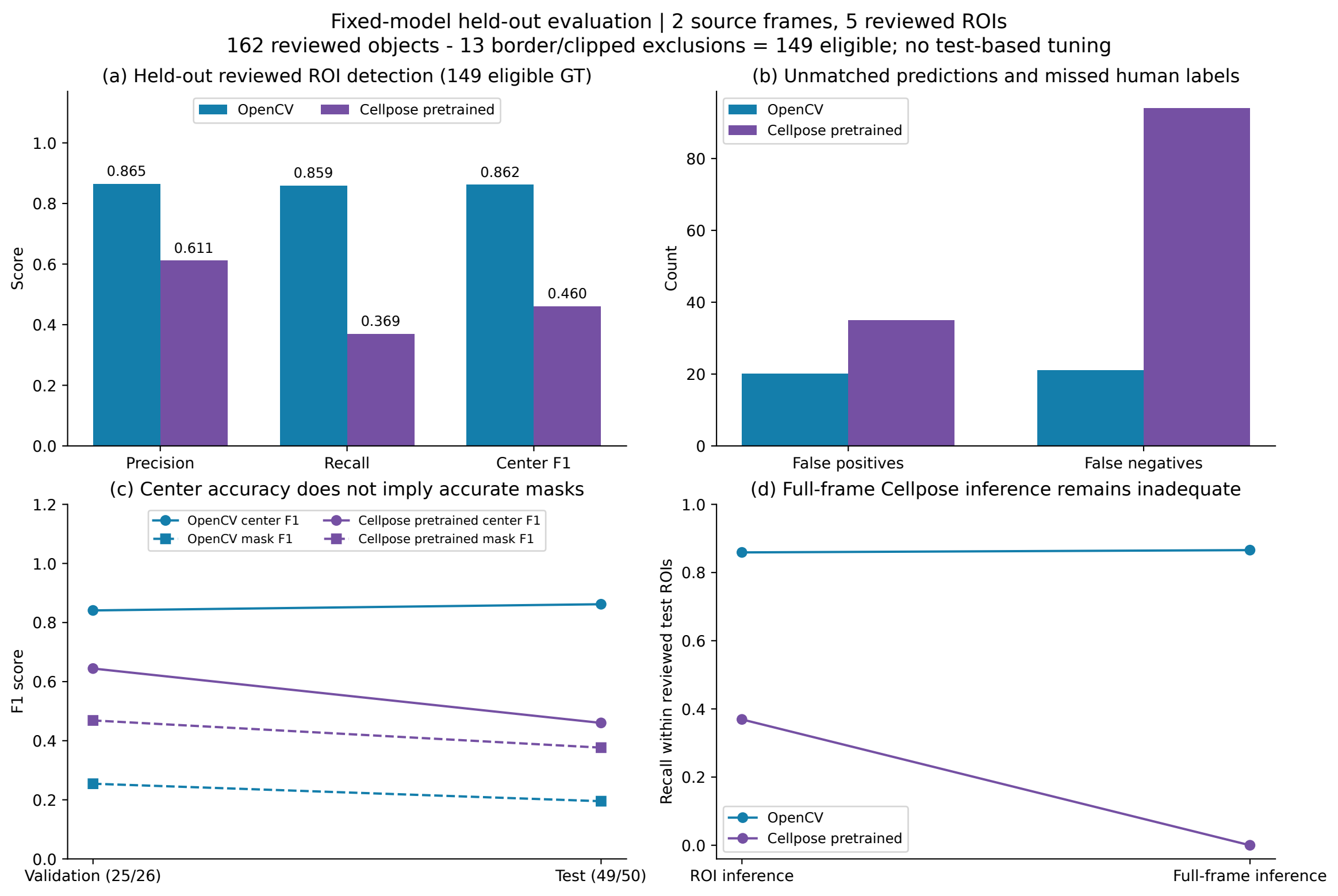
Fixed-model held-out evaluation | 2 source frames, 5 reviewed ROIs
162 reviewed objects - 13 border/clipped exclusions = 149 eligible; no test-based tuning
(a) Held-out reviewed ROI detection (149 eligible GT)
OpenCV
Cellpose pretrained
Score
0.865
0.611
0.859
0.369
0.862
0.460
Precision
Recall
Center F1
(b) Unmatched predictions and missed human labels
Count
False positives
False negatives
(c) Center accuracy does not imply accurate masks
OpenCV center F1
OpenCV mask F1
Cellpose pretrained center F1
Cellpose pretrained mask F1
F1 score
Validation (25/26)
Test (49/50)
(d) Full-frame Cellpose inference remains inadequate
Recall within reviewed test ROIs
ROI inference
Full-frame inference